

Youssef Khemiri

AI Engineer / Master's in Data science

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Professional Profile

AI Engineer with 3+ years of experience building AI solutions, data pipelines, production ready APIs, NLP/GenAI and Computer Vision systems. Actively involved in applied research, experimentation, and model evaluation, with a proven track record of delivering scalable systems that drive productivity and business impact.

Focus areas: Python, Machine Learning, Natural Language Processing, GenerativeAI, Computer Vision, Data Engineering, MLOps, AWS

Work History

03/2023 – 12/2025

AI Engineer – Working student

42Robots – 42robots.ai | United States (Remote)

Custom built AI powered enterprise solutions

Tech stack: Python, FastAPI, AWS, GenerativeAI, LLMs, Transformers, Open Source, SQL (Postgre), NoSQL (Mongo), VectorDB (Milvus, Pinecone, Chroma), RAG, Git, GitHub Actions, Docker, Selenium

- Boosted partner operational throughput by 50% by building an end to end AI powered automation system.
- Automated large scale data acquisition and processing pipelines; Built and delivered AI Agents powered by LLMs, RAG, GCP APIs and custom integrations.
- Built and deployed FastAPI services on AWS EC2 with relational data managed on AWS RDS.
- Attended and contributed to bi-weekly brainstorming and research sessions to discuss new projects, ideas and architectures.

07/2025 – 08/2025

Data science Intern

Esprit Tech – esprit.tn | Ariana, Tunisia

Tech stack: Python, scikit-learn, LLMs, OpenAI API, Hugging Face, Transformers, Pytorch, MLflow

- Scraped and analyzed 1,400 real estate listings from the local market by building an automated scraper using BeautifulSoup and Selenium from 3 providers.
- Improved XGBoost performance by 10% over the baseline system by applying LLM based description analysis, feature engineered 13 new features. Experimented with GridSearch for multiple ML models.

06/2024 – 07/2024

AI Engineering Intern

Arrow International | Hammamet, Tunisia (Remote)

Tech stack: Python, RAG, LangChain, OpenAI, NLP, Guardrails

- Developed a Youtube LLM powered Agent enabling real time summarization, key point extraction, and Q&A over video transcripts.
- Applied NLP and guard rails techniques for cost efficient, context aware and safe conversations.

05/2023 – 06/2023

Python Instructor

GoMyCode – gomycode.com | Nabeul, Tunisia

- Delivered an applied Python, data structures, Numpy, Pandas and web scraping course with structured workshops and code reviews.
- Achieved a 5 star review from students

- 02/2023 – 06/2023

Data science intern
For Right Solutions (FRS) – frsdev.com | Sousse, tunisia
Tech stack: Python, TensorFlow, Matplotlib, Seaborn, Pandas, Numpy, Streamlit, CNN,
 - Built an end-to-end image classification platform using Python, Streamlit, and TensorFlow.
 - Implemented large data pre-processing, custom CNNs, transfer learning, and evaluation dashboards.
- 11/2022 – 05/2023

Python and AI Freelancer
UpWork – upwork.com | Remote
Global online freelance marketplace
Tech stack: Python, TensorFlow, Pytorch, Hugging Face, scikit-learn, Matplotlib, Seaborn, Pandas, Numpy, Streamlit, RAG, Langchain, finetune
 - Achieved Top-rated freelancer badge by landing 15 contracts, 800+ hours of work, and 100% Job Success score.
 - Boosted app speed by 30% by building an optimized data loading process.
 - Reproduced a DL scientific research with 10% accuracy improvement over the original paper.
 - Delivered consistent daily research about AI papers and news for 277 days.

More academic and personal projects in AI and Data science can be found in my personal Portfolio.

Education

- 10/2025 – 03/2026

Master's in Data Science – Exchange program
Philipps Universität Marburg | Germany
- 09/2023 – 06/2026

Engineering Degree in Data Science & Software Engineering
ESPRIT Engineering | Tunisia
- 09/2020 – 06/2023

Bachelor in Computer Science
ISSAT Sousse | Tunisia
- 09/2018 – 06/2020

Baccalaureate in Mathematics
LMBH, Hammamet | Tunisia

Additional Information

Languages

- English (C1), Arabic (Mother Tongue), French (B2), German (A2)

Involvement

- IEEE Member | ARSII Member | Former Technical Mentor (lablab.ai) | Former Chairman (IEEE ISSATso CS SBC) | Former Ambassador (IEEEExtreme 15.0) | Former member at Problem Solving Division (competitive programming)

Awards

- 1st Prize — MLM AI Competition (Generative AI, Open-Source LLMs)
- 2nd Place — IEEE TSYP National Competition (Computer Vision)
- Best Ambassador award — IEEE ENIG TUN'IM 2.0